

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE:ITA0604

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:100

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I D.RITHVIK (192312119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Total Marks Awarded

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	30%	20%	20%	15%	15%	
Q1						
Q2						
Q3						

Short Answer Test

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models. (5 Marks)

7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

8. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

6. Decision Tree Learning System

Architecture:

Input Data -> Feature Selection -> Decision Tree Induction -> Pruning -> Final Model -> Prediction.

Feature Selection:

- Entropy measures impurity.
- Information Gain = Entropy(before) – Entropy(after split).
- Choose the feature with highest information gain.

Example:

Weather (Sunny, Rainy), Play (Yes/No). If 'Weather' gives highest information gain, it becomes the root node. Continue splitting until leaf nodes are pure.

Pruning:

- Pre-pruning: stop early using depth/min samples.
- Post-pruning: remove unnecessary branches after training.

Overfitting Control:

- Limit tree depth.
- Increase minimum samples per split.
- Apply pruning.
- Use cross-validation.

7. Smart Traffic Management System:

Problem:

Predict traffic congestion using vehicle density, time, weather, and road type.

Algorithm:

Random Forest is suitable because it handles mixed features, nonlinear patterns,

and reduces overfitting.

Preprocessing:

- Handle missing values.
- Encode categorical features.
- Normalize if required.
- Remove outliers.

Feature Engineering:

- Peak/off-peak indicator.
- Rain intensity.
- Traffic density ratio.

Training & Validation:

- Split into training/testing sets.
- Train Random Forest.
- Validate using Accuracy, Precision, Recall, F1-score.

Deployment:

Collect real-time sensor/GPS data, predict congestion, and send alerts or optimize traffic signals.

8. Convolutional Neural Networks (CNNs)

Layers:

- Convolution Layer: extracts edges, textures, and patterns using filters.
- Pooling Layer: reduces feature map size and computation.
- Fully Connected Layer: performs final classification.

Feature Maps:

Filters slide over the image to produce activation (feature) maps.

Backpropagation:

Prediction error is calculated, gradients are computed, and weights are updated

using gradient descent.

Example:

Input image -> Convolution -> ReLU -> Pooling -> Convolution -> Pooling -> Fully Connected -> Softmax -> Output (Cat/Dog).

Factors Affecting Performance:

- Dataset quality and size.
- Number of layers and filters.
- Learning rate.
- Batch size.
- Regularization and data augmentation.